Teaching Package: Living Organisms and Non-Living Things Subject: Science | Grade: Grade 4 | Duration: 45 minutes | Language: English *Assumption: This lesson follows standard primary science curricula focusing on the biological characteristics of life.*

A. Learning Objectives and Success Criteria

Learning Objective: Students will be able to classify objects as living or non-living by identifying and applying the seven life processes (MRS GREN: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition).

Success Criteria (I Can Statements):

- I can list the things that all living organisms need to do to stay alive.
- I can sort items into "living" and "non-living" categories.
- I can explain why an object is living or non-living using evidence.

B. Teacher Explanation (teacher-facing)

Step-by-Step Explanation:

1. **The Hook:** Hold up a rock and a potted plant. Ask: "What makes these different?"
2. **The MRS GREN Mnemonic:** Introduce the acronym to help students remember the characteristics of life.
    - **M**ovement: Can it move on its own?
    - **R**espiration: Does it use energy from food/air?
    - **S**ensitivity: Does it react to its surroundings?
    - **G**rowth: Does it get bigger or change over time?
    - **R**eproduction: Can it make more of itself?
    - **E**xcretion: Does it get rid of waste?
    - **N**utrition: Does it need food or water?
3. **The Rule:** To be considered "living," an organism must demonstrate *all* of these characteristics at some point in its life cycle.

Essential Vocabulary:

- **Organism:** A living thing.
- **Non-living:** Something that was never alive or is made of dead material.
- **Characteristic:** A feature or quality that helps identify something.

Addressing Misconceptions:

- *Misconception:* "If it moves, it's alive." (e.g., a car or a cloud).
    - *Correction:* Use the checklist. A car moves, but does it grow? Does it reproduce? No.
- *Misconception:* "Plants aren't alive because they don't move around."

- *Correction:* Explain that plants move (turning toward sunlight) and perform all other functions, just much slower than animals.

Inclusive Strategies (UDL/Dyslexia Support):

- Provide a visual checklist (icons for each letter of MRS GREN).
- Use high-contrast text and sans-serif fonts (e.g., Arial or OpenDyslexic).
- Allow verbal sorting for students who struggle with handwriting.

C. Student Handout (student-facing)

Living or Non-Living?

Everything in our world is either **living** (an organism) or **non-living**. To be alive, an object must do seven special things. We use the code **MRS GREN** to remember them:

- **M**ovement: It can move its body.
- **R**espiration: It uses energy.
- **S**ensitivity: It reacts to things around it (like light or touch).
- **G**rowth: It gets bigger or changes.
- **R**eproduction: It can make babies or new life.
- **E**xcretion: It gets rid of waste.
- **N**utrition: It needs food or water.

Examples:

- **Living:** A dog, a sunflower, a mushroom, a human.
- **Non-living:** A rock, a toy car, a pencil, a cloud.

Your Task:

1. Look at the objects on your desk or in the classroom.
2. Pick three objects.
3. Use the MRS GREN checklist to decide if each one is living or non-living.
4. Write your answer in the table below.

Object Name	Is it Living? (Yes/No)	Why? (Pick one MRS GREN reason)
Example: Plant	Yes	It grows and needs water.

Quick Review:

- If something is non-living, it does **not** need to do all seven things.
- If you are unsure, ask: "Does it grow? Does it need food?" If the answer is no, it is likely non-living! *Assumption: The curriculum focuses on the biological criteria for life (MRS GREN concept: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition).*

D. Differentiated Learning Versions

1. Beginner (short explanations, familiar words, worked example, visual supports, hints, small steps, frequent checks for understanding)

Living things grow and need food. Non-living things do not. **Worked Example:** A dog is living because it eats food and moves. A rock is non-living because it does not grow or eat. **Task:** Look at these pictures: a tree, a toy car, a bird, and a spoon.

- Circle the **Living** things in green.
- Circle the **Non-living** things in red. *Hint: Ask yourself, "Does it need water to stay alive?" If yes, it is living!*

2. Intermediate (grade-level explanation, independent practice, mix of recall/comprehension/application, explain reasoning)

To be classified as "living," an organism must perform specific life processes like growing, reproducing, and responding to its environment. **Task:** Create a T-chart in your notebook. Sort the following items: *a seed, a cloud, a butterfly, a bicycle, a fire, and a mushroom.* **Reasoning:** Choose one item from your "Living" list and one from your "Non-living" list. Write one sentence for each explaining *why* it belongs in that category based on its observable characteristics.

3. Advanced / Challenge (problem-solving, analysis, investigation, extension; compare, justify, predict or design)

Some objects can be tricky to classify (e.g., a fire or a robot). Analyze the "life processes" (Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition). **Task:** Select a "tricky" object like a fire or a smartphone.

- Justify why it might *look* living but is actually non-living.
- Design a "Fair Test" observation checklist that someone could use to prove an unknown object is definitely alive. What 3 specific behaviors would you look for to confirm life?

Teacher Guidance (Inclusive Practices)

- **For Dyslexic Learners:** Use high-contrast print, sans-serif fonts (like Arial or Verdana), and avoid italics. Use bullet points rather than dense paragraphs.
- **UDL Application:** Provide a word bank with visual icons (e.g., a picture of a sandwich next to the word "Nutrition"). Allow students to express their understanding through drawings or verbal recording if written output is a barrier.
- **Check for Understanding:** During the 45-minute block, perform a "thumbs up/down" check after sorting the first two items to ensure students grasp the core difference before moving to independent work. *Assumption: As no specific curriculum standards were provided, this lesson is aligned with general primary science inquiry-based learning principles appropriate for Grade 4.*

E. Dyslexia-Friendly Learning Version

Vocabulary Preview and Word Bank (new words in small groups)

- **Group 1 (Action): Grow**, **Breathe**, **Move**.
- **Group 2 (Nature): Living**, **Non-living**.
- **Word Bank:** living | non-living | grow | breathe | object.

Short Explanation (concise sentences, short sections with descriptive headings, key terms in bold, not colour alone)

What is Living? A **living** thing is something that can **grow**, **breathe**, and make more of itself. Examples: a dog, a tree, or a person.

What is Non-living? A **non-living** thing does not grow or breathe. It cannot make more of itself. Examples: a rock, a toy, or a chair.

Guided Example (before independent work)

Look at a plant.

1. Does it **grow**? Yes.
2. Does it **breathe**? Yes.
3. Is it **living**? Yes.

Look at a pencil.

1. Does it **grow**? No.
2. Does it **breathe**? No.
3. Is it **living**? No.

Student Tasks (numbered sequential steps, short task sections, sentence starters)

1. Pick one object from your desk.
2. Ask: Does it **grow**? Does it **breathe**?
3. Place it in the **living** pile or the **non-living** pile.
4. Finish this sentence: "The ___ is **living** because it can ___."
5. Finish this sentence: "The ___ is **non-living** because it cannot ___."

Ways to Show Understanding (oral answers, diagrams, sorting, practical demonstration)

- **Sorting:** Physically move objects into two labeled boxes.
- **Diagrams:** Draw a circle around living things and a square around non-living things in a provided image.
- **Oral:** Explain your choice to a partner using the provided sentence starters.

Adaptations Used (extra processing time, text-to-speech or audio, no penalty for spelling or decoding errors, clear layout, sans-serif font in the final document, and the purpose of each)

- **Sans-serif font (e.g., Arial or Verdana):** Reduces visual crowding of letters.
- **Chunking information:** Breaking text into small, manageable sections prevents cognitive overload.
- **Sentence starters:** Reduces the barrier of writing production so students can focus on the scientific concept.
- **No penalty for spelling/decoding:** Ensures assessment measures scientific understanding rather than literacy proficiency.
- **Visual cues (boxes/piles):** Provides a concrete way to process the abstract concept of classification.
- **Extra processing time:** Allows students with slower auditory or visual processing speeds to internalize the vocabulary. *Assumption: As no curriculum standards were provided, this lesson focuses on foundational scientific literacy regarding biological classification based on the MRS GREN life processes (Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition).*

F. Examples and Activities

3 Age-Appropriate Examples:

1. **A Potted Sunflower:** Living (Needs water/sun, grows, reproduces).
2. **A River Rock:** Non-living (Does not grow, does not need food, cannot reproduce).
3. **A Toy Robot:** Non-living (It moves, but it does not grow or need nutrition; it requires batteries/external power).

Guided and Independent Practice:

- **Guided:** As a class, use a "Living vs. Non-living" T-chart on the whiteboard. Sort images of a mushroom, a bicycle, and a cloud together.
- **Independent:** Students receive a "Science Detective Sheet" where they must categorize 10 items (e.g., puppy, pencil, coral, feather) and write one reason for each choice.

Interactive/Hands-on Activity:

- **"Nature Walk Sort":** Take students outside or provide trays of collected items (leaves, twigs, pebbles, plastic bits). Students work in pairs to place items into two hoops labeled "Living" and "Non-living."

Visual/Multisensory Activity:

- **"The Life Card Shuffle":** Provide cards with pictures of life processes (e.g., a baby eating, a seed sprouting). Students hold the card and act out the action, then decide if the action proves the object is "Living."

Adaptation Notes:

- **Beginner:** Provide a visual word bank with icons next to the MRS GREN criteria.
- **Intermediate:** Students work in pairs to write one sentence per item.
- **Advanced:** Ask students to identify "borderline" items (e.g., a dried seed) and argue why it is living despite appearing dormant.
- **Dyslexia-Friendly:** Use sans-serif, high-contrast fonts. Provide printed sorting cards with clear, isolated images (avoid cluttered backgrounds). Allow oral responses instead of writing.

G. Practice Exercises

1. Multiple-Choice: Which of these is a sign that something is alive? a) It is shiny. b) It can grow. c) It is found in a house.

2. Matching: Match the item to its correct category.

- A Cloud — [ ] Living
- A Frog — [ ] Non-living
- A Car — [ ] Non-living
- A Tree — [ ] Living

3. Fill-in-the-Blank: Use the word bank: (Growth, Non-living, Nutrition, Living)

- A dog needs food to get __.
- A rock does not change size; it is __.
- __ things need energy, air, and water to survive.
- When a baby animal gets bigger, this is called __.

4. Short Answer: Why is a toy car not considered a "living thing" even though it can move?

5. Application: You find an object in your garden. It is green, has roots, and is getting taller. Is it living or non-living? List two clues that helped you decide.

H. Discussion Questions

- **Comprehension:** What are the basic things that all living organisms need to stay alive?
- **Explain your thinking:** If a robot can move and talk, why do we still classify it as non-living?
- **Real-life application:** If you find a dried-up leaf on the sidewalk, is it still "living"? Why or why not?
- **Extension (Advanced):** Can you think of an object that is "non-living" now but was once "living"? (e.g., a wooden chair). Explain how it changed.

Sentence Starters:

- "I think [item] is living because it ."
- "The difference between the two is ."
- "Even though it moves, it is not living because ."

Oral-Response Alternatives:

- Use "thumbs up/thumbs down" cards to answer whether an object is living.
- Allow students to record their answers on a digital device rather than writing them. *Assumption: This lesson aligns with general primary science inquiry-based standards for classifying biological and physical objects.*

I. Quiz and Answer Key

Audience: Students (Grade 4) | Level: Emerging/Developing

Name: _

Directions: Read the sentences and look at the images (or listen to your teacher read the questions). You may circle your answer, draw a line, or point to your choice.

1. **Circle the living things:** *Cat — Rock — Sunflower — Toy Car — Puppy*
2. **True or False:** All living things need energy (food/sunlight) to survive. (Circle one) **TRUE / FALSE**
3. **Fill in the blank:** A plant is a living thing because it can _ (grow / stay the same).
4. **Matching:** Draw a line from the word to the correct group.

- Cloud | Living
- Bee | Non-Living

1. **Short Answer:** Name one non-living thing in this classroom and tell me one reason why it is not alive.

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6. **Sorting:** Imagine you find a mushroom on a tree. Is it living or non-living? (Circle one) **LIVING / NON-LIVING**

Answer Key (teacher-facing)

Audience: Teacher

- **1. Correct:** Cat, Sunflower, Puppy. (Reasoning: These are biological organisms that carry out life processes).

- **2. Correct:** True. (Reasoning: Living things require energy for growth and repair).
- **3. Correct:** Grow. (Reasoning: Growth is a primary characteristic of living things).
- **4. Correct:** Cloud → Non-living; Bee → Living. (Reasoning: Bees respire, move, and grow; clouds are weather phenomena).
- **5. Correct:** Answers will vary (e.g., "Desk - it does not eat or grow").
- **6. Correct:** Living. (Reasoning: Mushrooms are fungi; they grow, reproduce, and require nutrients).

Scoring Guide:

- 6/6: Meets expectations.
- 4-5/6: Developing proficiency.
- 0-3/6: Needs reteaching/small group support.

Accessible Presentation/Response Options:

- **Read Aloud:** Provide a scribe or read the quiz to students with reading challenges.
- **Oral Response:** Allow students to explain their reasoning verbally to the teacher.
- **Visual Supports:** Provide real photos or concrete objects during the quiz if the student struggles with abstract text.
- **UDL/Dyslexia Support:** Use Sans Serif fonts (e.g., Arial or Verdana), increase line spacing to 1.5, and ensure high contrast. Provide extra time for processing.

J. Summary

Audience: Students/Teachers

Lesson Summary: This lesson guides students to differentiate between living and non-living things by identifying key biological characteristics: movement, respiration, sensitivity, growth, reproduction, excretion, and nutrition (often remembered by the acronym MRS GREN).

Key Takeaways:

1. Living things grow, reproduce, and react to their environment.
2. Non-living things do not perform life processes and do not need food or water.
3. If an object was once part of a living thing (like a wooden pencil), it is now considered non-living.
4. Using observable evidence (e.g., "It moves," "It needs water") is the best way to classify objects.

Student-Friendly Recap: "Today we became scientists! We learned that living things are special because they grow and change. If it breathes, eats, or grows, it’s alive. If it doesn't need those things to survive, it's non-living. Look around the room—can you spot three of each?"

Optional Vocabulary Review:

- **Organism:** A living thing that carries out all basic life processes.
- **Characteristics:** The special features that help us tell things apart.
- **Classify:** To put things into groups based on how they are the same or different.
- **Non-living:** An object that does not grow, breathe, or reproduce. *Assumption: This lesson aligns with general primary science expectations for identifying characteristics of life (MRS GREN simplified: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition).*

K. Visual-Content Prompts

Note: For all visuals, ensure essential information is also presented as simple, bolded text to support literacy and cognitive processing.

1. **Labeled Diagram (The Living Plant vs. The Non-Living Rock):**

- Image: A side-by-side illustration of a potted sunflower and a smooth river stone.
- Labels: Use color-coded arrows. Sunflower: "Grows," "Needs Water," "Needs Sunlight." Rock: "Does not grow," "Does not need food."
- Visual Cue: Icons representing "Growth" (an arrow pointing up) and "Needs" (a water drop).

1. **Step-by-Step Concept Visual (How to Test for Life):**

- Three simple frames:
  - Frame 1: Does it grow or change size?
  - Frame 2: Does it need food or energy?
  - Frame 3: Does it make more of itself (reproduce)?
- Icons: A ruler (growth), a bowl/sun (energy), and a baby animal/seedling (reproduce).

1. **Sorting Cards (Living vs. Non-Living):**

- Layout: Two distinct boxes at the top labeled "LIVING" (Green border) and "NON-LIVING" (Grey border).
- Items: High-contrast, realistic images of a dog, a bicycle, a mushroom, a puddle, a bird, and a ball.
- Design: Use large, sans-serif fonts (e.g., Arial or Lexend) for all labels.

1. **Simple Infographic (Living Things Checklist):**

- Heading: "Is it Alive?"
- Content: A checklist format. "A living thing must do ALL of these: [ ] Needs food/water, [ ] Grows, [ ] Can have babies, [ ] Reacts to the world."
- Summary: A concluding "YES" or "NO" box at the bottom.

L. Accessibility and Teacher Guidance (teacher-facing)

The adaptations used in each version:

- **Visuals:** Reduced background noise (white space) to prevent visual crowding.
- **Fonts:** Use of dyslexia-friendly fonts (e.g., OpenDyslexic or Lexend).
- **Color Coding:** Consistent use of green for "Living" and grey for "Non-living" throughout all materials to reinforce categorization patterns.

The purpose of each dyslexia-support strategy:

- **Text-to-Visual Pairing:** Reduces the cognitive load of decoding text by providing immediate visual context.
- **Graphic Organizers:** Helps students organize information spatially rather than relying solely on sequencing sentences, which is often difficult for dyslexic learners.
- **Bold/Sans-Serif Fonts:** Minimizes letter-flipping and visual distortion (e.g., 'd' vs 'b').

Ways to monitor student understanding:

- **"Think-Pair-Share" Observation:** Listen for students using the specific characteristics (e.g., "It's living *because* it needs water") rather than just guessing based on appearance.
- **Sorting Check:** During the sorting task, ask students to explain *one* choice, confirming they are applying a rule rather than relying on intuition.

Options for extra scaffolding or extension:

- **Scaffolding:** Provide a "Word Bank" with visual icons for students who struggle with written output. Offer sentence stems: "The ___ is living because it ____."
- **Extension:** Challenge high-performing students to identify "tricky" items, like a dry seed or a piece of wood, and justify why they might be categorized differently based on whether they are dormant or currently functioning.

Limitations and Verification:

- **Assumptions:** I have assumed the classroom has basic supplies for sorting. If technology is used, ensure images are not pixelated, as this hinders recognition for students with visual processing difficulties.
- **Standards Alignment:** This content is designed to support inquiry-based learning. **I have not aligned this to a specific state or national curriculum.** You must verify that the "Living vs. Non-living" classification criteria match your local district's specific learning targets (e.g., some curricula introduce "Respiration," while others focus strictly on "Growth").
- **Accuracy Check:** Ensure the "Non-living" items chosen do not imply they were never alive (e.g., a wooden chair was once part of a living tree), as this can confuse younger learners. Stick to clearly inanimate objects like stones, metal, or plastic for introductory lessons.